

CURRICULUM VITAE

Hao-Jui Kuan

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Professional experience

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| 2025 – Present | Research scholar at Illinois Center for Advanced Studies of the Universe & University of Illinois Urbana-Champaign, United States. <i>PI: Nicolas Yunes, Jacquelyn Noronha-Hostler.</i> |
| 2023 – 2025 | Postdoc in the Computational Relativistic Astrophysics division at the Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Germany. <i>PI: Masaru Shibata.</i> |

Education

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| 2019 – 2022 | University of Tübingen, Tübingen, Germany. <i>Advisor: Kostas D. Kokkotas</i>
PhD in Theoretical Astrophysics. Summa Cum Laude (excellent; $\lesssim 1$ in 50 graduates). |
| 2017 – 2022 | National Tsing Hua University, Taiwan. <i>Advisor: Chao-Qiang Geng</i>
PhD in Physics (joint degree). |
| 2013 – 2017 | National Tsing Hua University, Taiwan.
BSc Double major in Physics and Mathematics. |

Honors and Awards

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| 2023 | Honorable Mention in the GWIC- Braccini Thesis Prize |
| 2023 | Dr. Friedrich Förster Prize, one awardee in Faculty of Science in University of Tübingen, €1,000. |
| 2023 | Student Outstanding Paper Award from NCTS (Taiwan), converted as ~ 620 USD. |

Fellowships and Grants

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| 2020 – 2021 | [PI] “Neutron stars as gravitational wave sources”, Sandwich-Scholarship Programme, DAAD and MOST (Taiwan), €13,500. |
| 2017 – 2020 | Presidential Scholarship of National Tsing Hua University (Taiwan), converted as $\sim 18,700$ USD. |

Publications

22 refereed + 4 preprints (13 first-author); h -index 13, 595 citations.
Full list: hao-jui.github.io/publication.pdf.

Reviewing Activities

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| Journals | Physical Review Letter, Physical Review D, International Journal of Modern Physics D, European Physical Journal C, Monthly Notices of the Royal Astronomical Society |
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List of Presentations

Plenary Talks and Colloquia

2025

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| Feb 11 | “Uncovering Extreme Matter and Gravity through Gravitational Waves from Neutron Star Mergers”, Academia Sinica, Taipei, Taiwan. |
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2022

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| Sep 10 | “Premerger Neutron Star Physics”, Eleventh Aegean summer school, Syros, Greece. |
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Invited Seminars/Conference Talks

2025

- Oct 8 “**Tidal Resonance: Nonlinearities revealed by numerical relativity**”, Astrophysics, Relativity, and Cosmology Seminar at UIUC.
- Jun 17 “**Tidal Resonance in Binary Neutron Star Mergers – Numerical Relativity Insights**”, [Conference website](#), Varna, Bulgaria.
- Feb 27 “**Numerical Relativity Insights into Tidal Resonance Phenomena**”, University of Tübingen, Germany.

2024

- Nov 25 “**Probing Subatomic Physics and Gravity through Gravitational Waves from Neutron Star Mergers**”, Academia Sinica, Taipei, Taiwan.
- Aug 1 “**On the finite-size imprints on waveforms of binary neutron star mergers**”, UIUC.
- Jul 11 “**Imprints of matter and scalar effects on waveforms from binary neutron star mergers**”, online-seminar-in-mathematical-numerical-relativity ([website](#)).
- Apr 24 “**Imprints of matter and scalar effects on waveforms from binary neutron star mergers**”, Institute for Gravitational and Subatomic Physics (GRASP), Utrecht University.
- Apr 8 “**(Dynamical) Tidal Effects in the Binary Neutron Star mergers**”, AstroParticle and Cosmology laboratory (APC), Paris, France.

2023

- Mar 16 “**Dynamical Scalarization during Neutron Star Mergers in scalar-Gauss-Bonnet Theory**”, University of Tübingen, Germany.
- Feb 3 “**Developing waveform involving dynamical tides**”, Academia Sinica, Taipei, Taiwan.
- Feb 2 “**Gravitational Phase Transition in Massive Scalar-tensor Theory**”, National Center for Theoretical Sciences, Hsinchu, Taiwan.

2021

- Jul 27 “ **g -mode resonances as triggers for precursors of SGRBs**”, Grav@zon group, Federal University of Pará, Pará, Brazil.
- Jun 22 “**Dynamical formation of scalarized black holes and neutron stars through stellar core collapse**”, cosmo/GW journal club, Johns Hopkins University, Maryland, USA.

Contributed Talks

2025

- Jul 24 “**Tidal Resonance in Binary Neutron Star Mergers**”, New Frontiers 2025 in Numerical Relativity, Mallorca, Spain.

2024

- Oct 9 “**On the finite-size imprints on waveforms of binary neutron star mergers**”, Observatoire de Paris - site de Meudon, France.
- Feb 28 “**Binary neutron star mergers in massive scalar-tensor theory**”, Gravity and Cosmology 2024, Yukawa Institute for Theoretical Physics, Kyoto, Japan.

2023

- Oct 26 “**Binary neutron star mergers in massive scalar-tensor theory: an adiabatic look**”, poster, GravityShapePisa 2023, Pisa, Italy.
- Apr 5 “**Dynamical Scalarization during BNS mergers in scalar-Gauss-Bonnet**”, CoCoNut meeting, Potsdam, Germany.
- Mar 22 “**Packed Message delivered by Tides in Binary Neutron Star Mergers**”, SMuK2023, Technical University Dresden, Germany.

2022

- Sep 8 “**Gravitational Phase Transition**”, Eleventh Aegean summer school, Syros, Greece.
- May 16–19 “**Tidal effects in the pre-merger stage of coalescing binary neutron stars**”, e-poster, PHAROS: The multi-messenger physics and astrophysics of neutron stars, Roma, Italy.
- Mar 3 “**Resonance shattering as triggers for precursors of SGRBs**”, DPG Meeting of the Matter and Cosmos section (SMuK), Heidelberg, Germany.

2021

- Aug 30 – Sep 3 “**Tidal g -mode resonances in coalescing binaries of neutron stars as triggers for**

precursor flares of short gamma-ray bursts”,
DPG Meeting of the Matter and Cosmos section (SMuK), Bad Honnef, Germany.
Feb 24–28 “Inverse-Chirp Imprint of GW in Scalar Tensor Theory”,
56th Karpacz Winter School in Theoretical Physics, Karpacz, Poland.

Advising and Mentoring

Current Graduate Students

2025 – Present Carlos Conde, UIUC, co-advised with Prof. Yunes.
2025 – Present Daniel Caballero, UIUC, co-advised with Prof. Yunes.
2025 – Present Zoey Dong, UIUC, co-advised with Prof. Yunes.

Teaching Activities

Teaching assistant

2016 – 2019 *Tsing Hua*: General Relativity (2016, 2018 Fall), Mathematical Physics (2017–18),
Thermal Physics (2016–17), Calculus (2018–19)
2021 – 2022 *Tübingen*: Physical Practical (2021 Summer & Winter), Grundkurs
Electromagnetism (2022 Summer)

Outreach

Mar 2025 Black hole lectures, *Jürgen Ehlers Spring School*, Potsdam.

Computer Skills and Languages

Programming Python, Fortran, C++
Scientific software Matlab, Mathematica, Gnuplot
Codes **GRASS** (General Relativistic Axisymmetric Spacetime Solver)
Kadath-AEI (AEI fork of Kadath library and FUKA API)